

Stage 1: Business Understanding & Data Collection

Stock Move Prediction
Fintech Workshop | Reichman University

Step 1: Data Description

- **Define your goal:** In a few sentences, state what you are trying to predict. The shared task is a binary classification will a stock in your assigned sector rise tomorrow.
- **Data Source:**
 - Daily OHLCV data from Polygon.io for all tickers in your assigned sector, covering the past four years.
 - VIX and S&P 500 daily.
- **High-level overview of the data:**
 - Report the number of unique tickers in your sector and the number of times each ticker occurs.
 - Report the date range and total number of trading days covered
- **Potential features:**
 - Engineered features - What signals can you derive from the OHLCV data that might have predictive power? Common variables include moving averages, rolling volatility, RSI, distance from a moving average, return windows, and volume ratios, across different time horizons. Think about what could help your model predict 1 day return and list your initial candidates, you will test and refine them later.
 - External sector data - What external variables can you use for your sector? E.g. Energy → oil prices, Real Estate → mortgage rates, Tech → semiconductor index. Identify external data you plan to integrate, where you'll source it from, and why they should carry predictive power for your sector specifically.
- **Choose your threshold.** The default is 1%. If you choose a different threshold for your sector, explain why.
- **Construct the binary label.** Apply your threshold to the next-day return column: 1 if the return exceeds your threshold, otherwise 0. Report the number of positive observations as a percentage of all observations.

Output

1. **Written section (1–2 pages).** Covering the items above.